S4 Supplementary Clinical Summary

A brief, cautious clinician-facing note on this mechanistic/methods-level finding

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Short title: S4 Clinical Summary - RVC_SLEEP_peak_analysis

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Important: not yet a clinical finding

This article reports a basic-science, mechanistic/methods-level result about the timing of a well-known EEG event (the sleep spindle) relative to breathing. It tested **no** clinical population, intervention, or outcome. The single dataset (RichSleep, 24 subjects) is healthy and unselected. Nothing here supports any diagnostic, monitoring, or intervention use.

What was found, in plain language

Sleep spindles - brief bursts of brain activity during light-to-moderate sleep - tend to reach their **strongest** moment at a particular point in the breathing cycle (roughly a third of the way toward the top of an inhale), consistently across 24 healthy adults and several independent checks. This is a refinement of a companion study's earlier finding that the **start** of a spindle shows no such pattern - it turns out the timing signal is there, but only shows up when looking at the spindle's peak rather than its beginning. A weaker, less consistent version of the same idea was found for heartbeat timing. A separate, more tentative analysis found a hint - not yet a solid finding - that spindles occurring closer to this preferred breathing moment might also produce a slightly larger effect on the brain's own state-space geometry (the same geometric measure used in the companion study), but this needs independent confirmation before it should be trusted.

Why this is not (yet) clinically actionable

- This is a **timing** finding about when, within a spindle, brain and breathing activity align - not a finding about memory, cognition, sleep quality, or any clinical outcome. None of those were measured or tested here.
- The single dataset is healthy, unselected, and observational; no sleep disorder, cognitive condition, or intervention was studied.
- The secondary, more clinically-adjacent question (does this timing predict anything about the spindle's own downstream effect?) returned only a fragile, partially-inconsistent hint (2 of 18 statistical tests, with mixed agreement across follow-up checks) - explicitly not strong enough to build on without independent replication.
- "Statistically significant" in this article, as elsewhere in this research program, is a multiple-comparison-corrected statistical statement, not a claim about clinical meaningfulness or effect size at the individual level.

What would make this more clinically relevant

- Replicating the respiratory peak-phase timing effect in an independent, ideally larger and/or clinical (e.g. sleep-apnea, insomnia) PSG sample.
- A dedicated, pre-registered replication of the fragile secondary (alignment-moderation) result, ideally with a genuine endogenous spindle-strength covariate (e.g. slow-oscillation-spindle coupling), which is not available in the present data.
- Any demonstration that this timing relationship, or its downstream geometric consequence, differs between healthy and clinical populations - entirely untested here.

Summary for clinical readers

This is a basic-science refinement of exactly **where within a sleep spindle** a breathing-related timing signal appears (its peak, not its onset), replicated across four independent robustness checks in one healthy 24-subject dataset. A secondary, much more tentative hint that this timing might also relate to the spindle's downstream brain-state effect is reported honestly as fragile and unconfirmed. Neither result currently supports any diagnostic, monitoring, or intervention use.

Bibliography